



Unit 1 — Basic concepts of Artificial Intelligence

CHAPTER 1.2 • DATA, LOGIC & COMPUTING

Role of Data & Information and the Evolution of Computing

Types of Data • Identifying & Acquiring Data • Binary Logic System

Conditional Gates • Deterministic & Probabilistic Systems

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What We Will Cover



01

Role of Data & Information

Why data matters, and how it becomes information



02

Types of Data

Structured, unstructured and semi-structured data



03

Identifying, Acquiring & Exploring Data

From raw sources to actionable insight



04

Binary Logic System & Conditional Gates

Binary numbers, logic gates and truth tables



05

Deterministic vs Probabilistic Problems

Real-life examples across weather, medicine and AI

Role of Data & Information



Data

Raw, unprocessed facts and figures with no context on their own — numbers, words, signals, or measurements collected from the real world.

Example: 36, 40, 25, 92 (a set of temperature readings)



Information

Data that has been processed, organised and given context so it becomes meaningful and useful for decision-making.

Example: "City temperature averaged 40°C this week — a heatwave alert is needed."

Every modern system — from a smartphone app to a hospital ICU — depends on turning raw data into trustworthy information for decisions.

Types of Data



Structured Data

Organised in rows & columns, fits neatly into databases and spreadsheets.

EXAMPLES

Bank ledgers, Excel sheets, SQL tables



Semi-structured Data

Has some organisational tags or markers, but no rigid table format.

EXAMPLES

JSON files, XML, emails with headers



Unstructured Data

No predefined format or organisation — the hardest to search and analyse directly.

EXAMPLES

Photos, videos, social media posts

Identifying Relevant Data



Define the Objective

Start with the question or problem to solve — it determines exactly what data is needed.



Check Relevance

Keep only data directly connected to the goal; discard data that adds noise, not value.



Assess Reliability

Prefer data from credible, verifiable sources over unverified or outdated ones.



Confirm Completeness

Ensure the data covers the full picture needed — missing fields weaken conclusions.



Real-life Example

A hospital predicting patient risk needs vitals, history and lab results — but not the patient's favourite colour. Identifying relevant data avoids wasted effort and bias.

Acquiring Data from Multiple Sources



Sensors & IoT

Temperature, motion, GPS and industrial sensors stream continuous readings.



Surveys & Forms

Direct responses collected from people — census, feedback, questionnaires.



Databases & Web

Structured records from company systems, APIs and public web sources.



Transactions

Purchase logs, banking records and point-of-sale data captured in real time.



Social Media & Apps

Posts, reviews, clicks and usage patterns from digital platforms.



Third-party Sources

Licensed datasets, government open data and partner data-sharing feeds.

Exploring & Visualising Data



Collect



Clean



Explore



Visualise



Insight

Why Exploration Matters

- Summary statistics (mean, min, max) reveal the overall shape of the data.
- Charts and graphs expose trends, outliers and patterns invisible in raw numbers.
- Example: plotting daily sales instantly shows a weekend spike a spreadsheet of numbers would hide.

The Binary Logic System

Digital computers represent every instruction, number, letter and image using just two states: 0 and 1 (Base-2). These correspond physically to low/high voltage, off/on switches, or magnetised/non-magnetised bits.

- Every character, image and sound is ultimately stored as combinations of 0s and 1s.
- Binary underlies logic gates, processors, memory and all digital circuits.
- Programs make decisions using binary logic — conditions are either True (1) or False (0).



Decimal → Binary

Decimal	Binary (Base-2)
0	0000
1	0001
5	0101
9	1001
13	1101

Each position represents a power of 2 (... 8, 4, 2, 1) — the same way decimal positions represent powers of 10.

Logic Gates at a Glance

AND

$$A \bullet B$$

Output is 1 only when all inputs are 1.

OR

$$A + B$$

Output is 1 when at least one input is 1.

NOT

$$\bar{A}$$

Reverses the input — the only single-input gate.

NAND

$$(A \bullet B)$$

AND followed by NOT — a universal gate.

NOR

$$(A + B)$$

OR followed by NOT — a universal gate.

XOR

$$A \oplus B$$

Output is 1 only when inputs differ.

AND & OR Gates in Detail

AND $A \cdot B$

Produces a 1 (True) output only when both inputs are 1 — every condition must be satisfied.

A	B	Out
0	0	0
0	1	0
1	0	0
1	1	1

EXAMPLES

- Alarm sounds only if motion sensor AND door sensor both trigger.
- Seatbelt warning lights only when engine runs AND belt is unfastened.

Rule: Multiplication-style logic — all conditions matter equally.

OR $A + B$

Produces a 1 (True) output when at least one input is 1 — any one condition is enough.

A	B	Out
0	0	0
0	1	1
1	0	1
1	1	1

EXAMPLES

- Streetlight turns on if darkness sensor OR manual switch is triggered.
- Fire alarm activates if smoke is detected OR the button is pressed.

Rule: Multiplication-style logic — all conditions matter equally.

NOT, NAND, NOR & XOR Gates

NOT A

Reverses a single input: 0 becomes 1, and 1 becomes 0.

A	Out
0	1
1	0

Light switch: if OFF → turn ON, and vice versa.

NAND $(A \bullet B)$

Opposite of AND — output is 0 only when both inputs are 1.

A	B	Out
0	0	1
1	1	0

Emergency stop: releases only when both buttons are held.

NOR $(A + B)$

Opposite of OR — output is 1 only when both inputs are 0.

A	B	Out
0	0	1
1	0	0

Standby mode: active only when no buttons are pressed.

XOR $A \oplus B$

Output is 1 only when the two inputs are different from each other.

A	B	Out
0	1	1
1	1	0

Staircase light: switch at either end toggles the same light.



NAND and NOR are called Universal Gates — any other logic gate (AND, OR, NOT, XOR) can be built by combining them, which is why they are the standard building blocks used inside real microchips.

Deterministic Systems



A deterministic system always produces the exact same output for a given input, with no randomness — the process and outcome are fully predictable and repeatable.

Characteristics

- Same input always gives the same output
- Fixed, non-adaptive process
- No randomness — fully predictable

Examples

- Sorting a list of numbers
- Designing a digital logic circuit
- Calculating compound interest

Limitation

- Cannot handle uncertainty
- Struggles in dynamic, changing environments
- Not suited to noisy real-world data

Probabilistic Systems



A probabilistic system deals with uncertainty and incomplete information, using statistical inference to estimate the most likely outcome rather than guaranteeing one exact answer.

Characteristics

- Handles uncertainty and incomplete data
- Relies on statistical inference
- Adaptable and flexible over time

Examples

- Weather forecasting
- Fraud detection in banking
- Spam email filtering

Strength

- Learns and improves from new data
- Works well with noisy, real-world inputs
- Powers modern AI and machine learning

Deterministic vs Probabilistic

Aspect	Deterministic	Probabilistic
Output for same input	Always identical	May vary; expressed as likelihood
Handles uncertainty?	No	Yes
Adaptability	Fixed process	Learns and adapts
Typical use	Sorting, circuit logic, arithmetic	Forecasting, diagnosis, fraud detection
Real-life example	A calculator computing 12×8	A weather app predicting rain tomorrow

Both Approaches at Work in Real Life



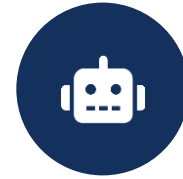
Weather Forecasting

Physics equations model air pressure deterministically, while probability estimates the chance of rain given uncertain atmospheric data.



Medicine & Diagnosis

Lab tests give deterministic values (e.g. blood sugar level), but diagnosing a disease often relies on probabilistic risk scoring.



Artificial Intelligence

Rule-based logic circuits are deterministic, while machine learning models make probabilistic predictions from patterns in data.

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Thank You

Questions & Discussion

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